

Déploiement et intégration d'un honeypot multi-services (Cowrie, Dionaea) avec Wazuh SIEM pour la détection d'intrusions et l'analyse des menaces

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Ce rapport présente une démonstration complète ainsi qu'un guide d'installation et de configuration pour la mise en place d'une architecture de sécurité basée sur :

- Wazuh (plateforme SIEM/IDS/XDR open source)
- Honeypots (Cowrie pour SSH/Telnet, Dionaea pour services SMB)
- pfSense (pare-feu et routage)

L'objectif est de simuler un environnement de détection d'intrusions où les attaques sont capturées, analysées et remontées dans Wazuh pour surveillance centralisée.

Présentation des technologies : Wazuh en tant que SIEM et Honeypots comme leurres de sécurité

WZUH :

Wazuh est une plateforme de sécurité open source de type SIEM (Security Information and Event Management) et XDR (Extended Detection and Response) qui fonctionne sur une architecture client-serveur avec des agents légers déployés sur les systèmes monitorés.

Architecture Technique de Wazuh :

- **Agent** : Collecte
- **Manager** : Traitement et analyse
- **Indexer** : Stockage et indexation
- **Dashboard** : Visualisation et reporting

HONEYPOT :

Un honeypot est un système informatique conçu pour attirer et détecter les activités malveillantes. Il sert principalement à surveiller les attaques, analyser les techniques des attaquants et collecter des informations sur les menaces. Les honeypots peuvent être simples (low-interaction) ou complexes (high-interaction) selon le niveau d'interaction avec les attaquants.

COWRIE :

Cowrie est un honeypot SSH et Telnet conçu pour simuler un serveur vulnérable.

Fonctionnalités principales :

- Simule un système Unix/Linux pour tromper les attaquants.
- Enregistre toutes les tentatives de connexion et les commandes exécutées.
- Permet de capturer les mots de passe et les commandes malveillantes pour analyse.
- Peut être intégré avec des outils de visualisation et de monitoring pour analyser les logs en temps réel.

Objectif :

Cowrie est surtout utilisé pour étudier les comportements des pirates sur les serveurs SSH/Telnet, comprendre les techniques de brute-force et identifier de nouvelles menaces.

DIONAEA :

Dionaea est un honeypot low-interaction spécialisé dans la capture de malwares.

Fonctionnalités principales :

- Simule différents services vulnérables (SMB, FTP, HTTP, etc.).
- Capture et stocke automatiquement les malwares envoyés par les attaquants.
- Permet l'analyse du malware dans un environnement sûr.
- Compatible avec des bases de données pour enregistrer toutes les attaques et malwares collectés.

Objectif :

Dionaea est utilisé pour collecter et analyser des malwares, ce qui permet aux chercheurs en sécurité de comprendre leur fonctionnement et leurs méthodes de propagation

Readme :

Configuration de PFSense :

```
Enter an option:

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 13523470316920bf9bb0

*** Welcome to pfSense 2.6.0-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd00::a00:27ff:fed9:605c/64
LAN (lan)      -> em1      -> v4: 192.168.1.1/24
OPT1 (opt1)    -> em2      -> v4: 172.16.0.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults   13) Update from console
5) Reboot system              14) Disable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █
```

```
Enter an option: 2

Available interfaces:

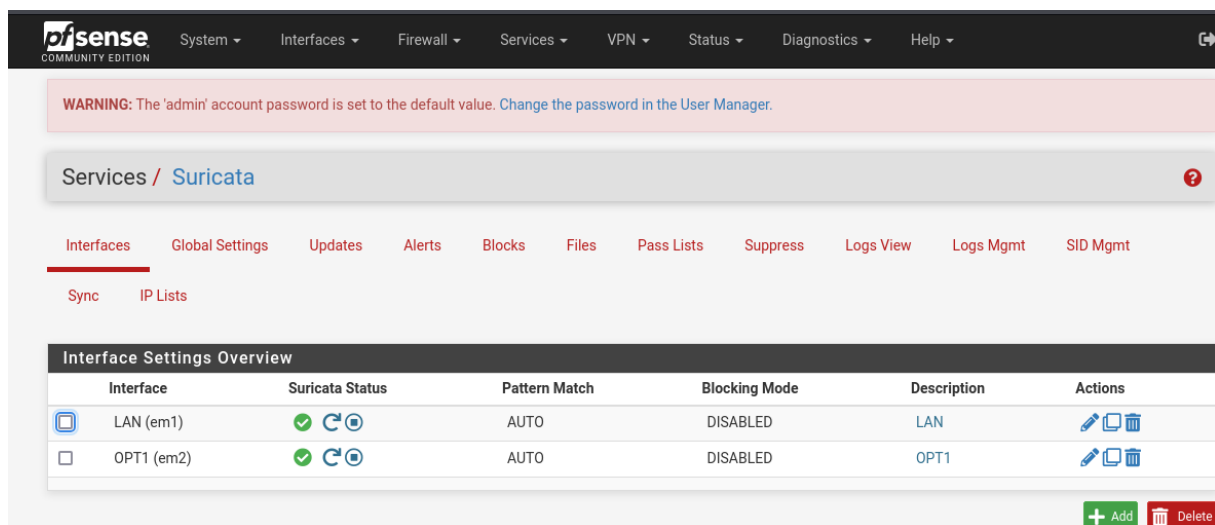
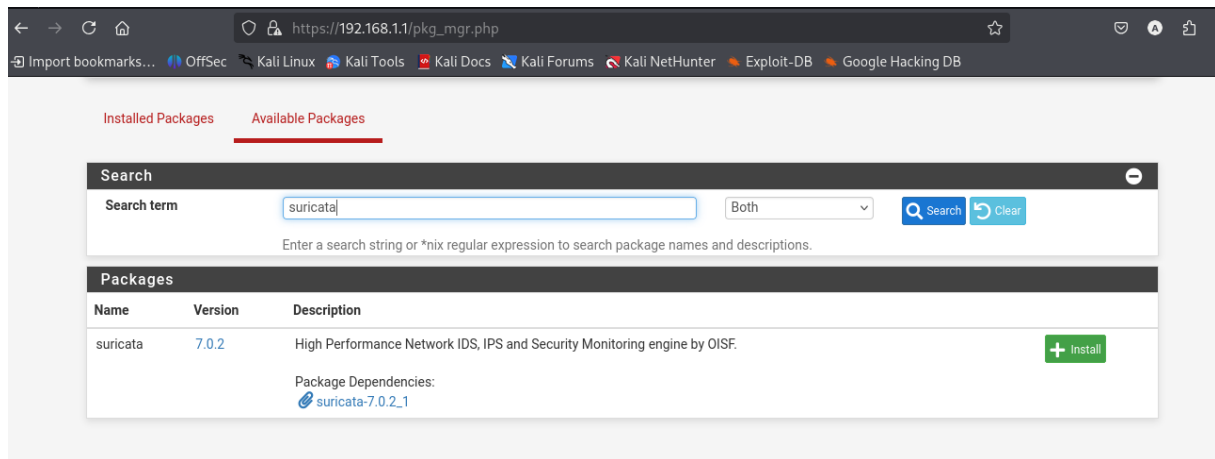
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)
3 - OPT1 (em2)

Enter the number of the interface you wish to configure: 3

Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 172.16.0.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24
```



Installation et configuration de WAZUH :

- Installation en mode All-in-One :

```
asmaa@asmaa:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
asmaa@asmaa:~$ chmod +x wazuh-install.sh
```

```
asmaa@asmaa:~$ sudo ./wazuh-install.sh -a
21/12/2025 01:56:39 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
21/12/2025 01:56:39 INFO: Verbose logging redirected to /var/log/wazuh-installer.log
```

```
21/12/2025 03:02:58 INFO: Wazuh dashboard web application initialized.
21/12/2025 03:02:58 INFO: --- Summary ---
21/12/2025 03:02:58 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: 3hCumR2wQxdJZQ4yUzoz6h4TSE?XfN1W
21/12/2025 03:02:58 INFO: Installation finished.
```

```
asmaa@asmaa:~$ sudo systemctl list-units --type=service | grep wazuh
wazuh-dashboard.service          loaded active running
g wazuh-dashboard
wazuh-indexer.service           loaded active running
g Wazuh-indexer
wazuh-manager.service           loaded active running
g Wazuh manager
```

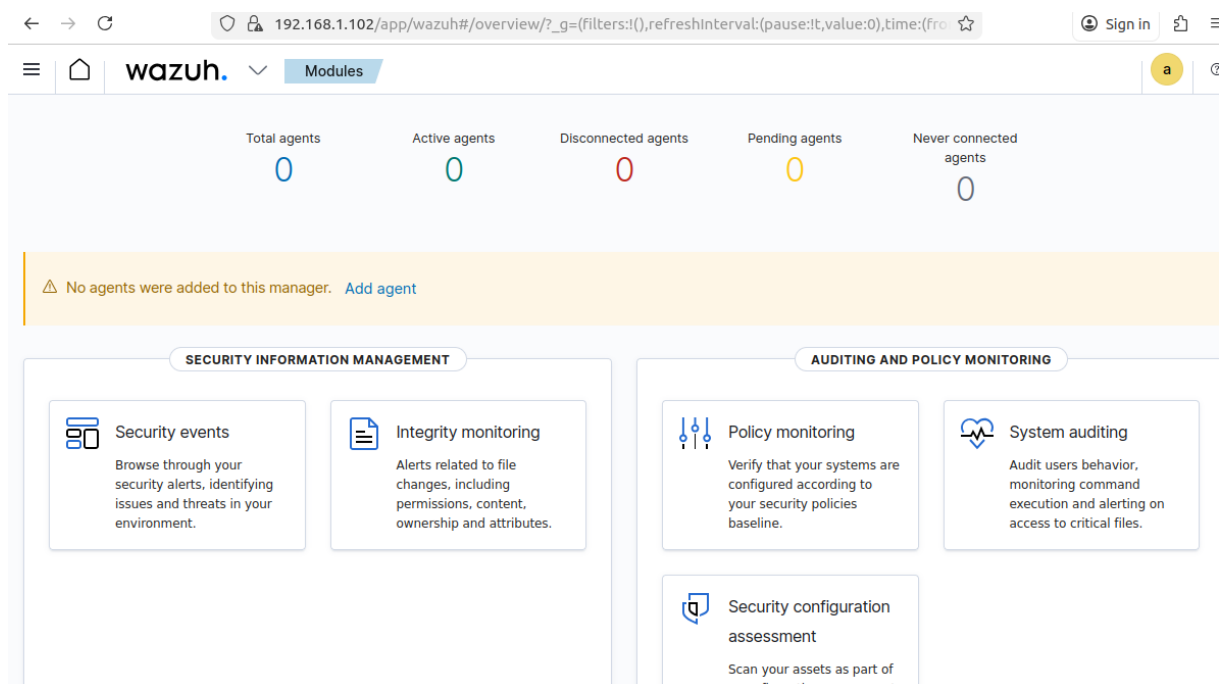
```
asmaa@asmaa:~$ systemctl start wazuh-dashboard
asmaa@asmaa:~$ systemctl start wazuh-manager
asmaa@asmaa:~$ systemctl start wazuh-indexer
```

```
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-23 10:10:47 +01; 5min ago
     Process: 4437 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 110 (limit: 2268)
   Memory: 25.4M
      CPU: 18.099s
```

```
asmaa@asmaa:~$ systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-23 10:02:56 +01; 14min ago
     Main PID: 561 (node)
        Tasks: 11 (limit: 2268)
       Memory: 185.1M
          CPU: 45.819s
      CGroup: /system.slice/wazuh-dashboard.service
              └─561 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-old-space-size=4096
```

```
asmaa@asmaa:~$ systemctl status wazuh-indexer
● wazuh-indexer.service - Wazuh-indexer
   Loaded: loaded (/lib/systemd/system/wazuh-indexer.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-23 10:16:00 +01; 2min 13s ago
     Docs: https://documentation.wazuh.com
     Main PID: 5092 (java)
        Tasks: 67 (limit: 2268)
       Memory: 1.1G
          CPU: 1min 52.630s
      CGroup: /system.slice/wazuh-indexer.service
              └─5092 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.disableFileShare=true
```

- Accès à l'interface web :



- Ajout d'agents :

✓

Select the package to download and install on your system:

 **LINUX**

☐ RPM amd64

☐ RPM aarch64

☒ DEB amd64

☐ DEB aarch64

 **WINDOWS**

☐ MSI 32/64 bits

 **macOS**

☐ Intel

☐ Apple silicon

Assign a server address: ?

192.168.1.102

✓

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: ?

Kali_Agent

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb &&  
sudo WAZUH_MANAGER='192.168.1.102' WAZUH_AGENT_GROUP='default' WAZUH_AGENT_NAME='kali_agent' dpkg -  
i ./wazuh-agent_4.7.5-1_amd64.deb
```

③ Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

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Start the agent:

```
sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

Déploiement des honeypots sur machine virtuelle (KALI) :

Installer Cowrie :

- Installer les dépendances :

`sudo apt install -y git python3 python3-venv python3-pip libssl-dev libffi-dev build-essential`

```
(kali㉿kali)-[~]
└─$ sudo apt install -y git python3-venv python3-pip libssl-dev libffi-dev build-essential
build-essential is already the newest version (12.12).
build-essential set to manually installed.
Upgrading:
  git                libpython3-dev      openssl-provider-legacy  python3-pip
  git-man            libpython3-stdlib    python3                  python3-pip-whl
  libffi-dev         libssl3t64          python3-dev              python3-venv
  libffi8            openssl              python3-minimal
Installing:
  libssl-dev
Suggested packages:
  libssl-doc
Summary:
  Upgrading: 15, Installing: 1, Removing: 0, Not Upgrading: 1442
  Download size: 21.8 MB
  Space needed: 16.6 MB / 61.7 GB available
```

- Créer un utilisateur :

Ne jamais lancer Cowrie en root

`sudo adduser cowrie`

`sudo usermod -aG sudo cowrie`

`su - cowrie`

```
(kali㉿kali)-[~]
└─$ sudo adduser cowrie
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for cowrie
Enter the new value, or press ENTER for the default
  Full Name []: cowrie
  Room Number []: cowrie
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y

(kali㉿kali)-[~]
└─$ sudo usermod -aG sudo cowrie
```



```
(kali㉿kali)-[~]  
$ su - cowrie  
Password:  
(cowrie㉿kali)-[~]  
$
```

- Télécharger Cowrie

```
git clone https://github.com/cowrie/cowrie.git  
cd cowrie
```

```
(cowrie㉿kali)-[~]  
$ git clone https://github.com/cowrie/cowrie.git  
Cloning into 'cowrie' ...  
remote: Enumerating objects: 20648, done.  
remote: Counting objects: 100% (987/987), done.  
remote: Compressing objects: 100% (493/493), done.  
Receiving objects: 0% (64/20648), 68.01 KiB | 26.00 KiB/s
```

```
(cowrie㉿kali)-[~]  
$ cd cowrie
```

- Créer l'environnement Python

```
python3 -m venv cowrie-env  
source cowrie-env/bin/activate  
pip install --upgrade pip  
pip install -r requirements.txt
```

```
(cowrie㉿kali)-[~/cowrie]  
$ python3 -m venv cowrie-env  
  
(cowrie㉿kali)-[~/cowrie]  
$ source cowrie-env/bin/activate  
  
(cowrie-env)(cowrie㉿kali)-[~/cowrie]  
$ pip install --upgrade pip  
Requirement already satisfied: pip in ./cowrie-env/lib/python3.13/site-packag  
es (25.3)  
  
(cowrie-env)(cowrie㉿kali)-[~/cowrie]  
$ pip install -r requirements.txt  
Collecting attrs==25.4.0 (from -r requirements.txt (line 1))
```

- Configuration de base de Cowrie

```
cp etc/cowrie.cfg.dist etc/cowrie.cfg  
nano etc/cowrie.cfg
```

Paramètres importants à vérifier :

[ssh]

enabled = true

listen_endpoints = tcp:2222:interface=0.0.0.0

[telnet]

enabled = true

listen_endpoints = tcp:2223:interface=0.0.0.0

Cowrie écoute sur 2222 (SSH) et 2223 (Telnet)

```
GNU nano 8.6          etc/cowrie.cfg *
#
[telnet]

# Enable Telnet support, disabled by default
enabled = true

# Endpoint to listen on for incoming Telnet connections.
# See https://twistedmatrix.com/documents/current/core/howto/endpoints.htm
# (default: listen_endpoints = tcp:2223:interface=0.0.0.0)
# (use systemd: endpoint for systemd activation)
# listen_endpoints = systemd:domain=INET:index=0
# For IPv4 and IPv6: listen_endpoints = tcp6:2223:interface=\:\: tcp:2223:
# Listening on multiple endpoints is supported with a single space separat
# e.g "listen_endpoints = tcp:2223:interface=0.0.0.0 tcp:2323:interface=0.
# use authbind for port numbers under 1024

listen_endpoints = tcp:2223:interface=0.0.0.0

GNU nano 8.6          etc/cowrie.cfg *
[ssh]

# Enable SSH support
# (default: true)
enabled = true

listen_endpoints = tcp:2222:interface=0.0.0.0
```

- Install Cowrie properly

Run from inside ~/cowrie:

```
(cowrie-env)(cowrie@kali)-[~/cowrie]
$ pip install -e .
Obtaining file:///home/cowrie/cowrie
Installing build dependencies ... \
```

- Lancer Cowrie

```
(cowrie-env)(cowrie@kali)-[~/cowrie]
$ cowrie start

Join the Cowrie community at: https://www.cowrie.org/slack/

Starting cowrie: [twistd --umask=0022 --pidfile /home/cowrie/cowrie/var/run/cowrie.pid --logger cowrie.python.logfile.logger cowrie] ...
/home/cowrie/cowrie/cowrie-env/lib/python3.13/site-packages/twisted/conch/ssh/transport.py:110: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-cbc": (algorithms.TripleDES, 24, modes.CBC),
/home/cowrie/cowrie/cowrie-env/lib/python3.13/site-packages/twisted/conch/ssh/transport.py:117: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-ctr": (algorithms.TripleDES, 24, modes.CTR),
```

- Vérifier état de cowrie:

cowrie status

Cowrie est actif

```
(cowrie-env)(cowrie@kali)-[~/cowrie]
$ cowrie status
cowrie is running (PID: 44488).
```

- Verify ports

```
(cowrie-env)(cowrie@kali)-[~/cowrie]
$ ss -tulnp | grep 2222
tcp    LISTEN 0      50      0.0.0.0:2222      0.0.0.0:*      users:(("twisted",pid=44488,fd=11))

(cowrie-env)(cowrie@kali)-[~/cowrie]
$ ss -tulnp | grep 2223
tcp    LISTEN 0      50      0.0.0.0:2223      0.0.0.0:*      users:(("twisted",pid=44488,fd=12))
```

Sur machine attaquant

- scanner la cible

nmap -p 22,23 192.168.1.104

```

(kali㉿kali)-[~]
$ nmap -p 2222,2223 192.168.1.104

Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-30 09:51 EST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.1.104
Host is up (0.0019s latency).

PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1
2223/tcp  open  rockwell-csp2
MAC Address: 08:00:27:9D:4A:47 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds

```

- Lancer attaque SSH :

```

(kali㉿kali)-[~]
$ ssh test@192.168.1.104 -p 2222

The authenticity of host '[192.168.1.104]:2222 ([192.168.1.104]:2222)' can't be established.
ED25519 key fingerprint is SHA256:JqW5iJ8qjpTycNvRI0M0w/l8/tLHGR4e+uY74g99/9c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.1.104]:2222' (ED25519) to the list of known hosts.
test@192.168.1.104's password:
Permission denied, please try again.
test@192.168.1.104's password:
Permission denied, please try again.
test@192.168.1.104's password:
test@192.168.1.104: Permission denied (publickey,password).

```

Retour sur la VM honeypot : Observer les attaques

tail -f var/log/cowrie/cowrie.log

Tu verras :

- IP attaquant
- Username / password
- Commandes
- URLs malveillantes

```

(cowrie-env)(cowrie@kali)-[~/cowrie]
$ tail -f var/log/cowrie/cowrie.log
2025-12-30T09:56:23.156486Z [HoneyPotSSHTransport,0,192.168.1.105] login attempt [b'test'/b'kali'] failed
2025-12-30T09:56:24.165890Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'test' failed auth b'password'
2025-12-30T09:56:24.166137Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized login: ()
2025-12-30T09:56:35.008321Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'test' trying auth b'password'
2025-12-30T09:56:35.010135Z [HoneyPotSSHTransport,0,192.168.1.105] Could not read etc/userdb.txt, default database activated
2025-12-30T09:56:35.010437Z [HoneyPotSSHTransport,0,192.168.1.105] login attempt [b'test'/b'asmaa'] failed
2025-12-30T09:56:36.012695Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'test' failed auth b'password'
2025-12-30T09:56:36.012989Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized login: ()
2025-12-30T09:56:36.017446Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2025-12-30T09:56:36.017706Z [HoneyPotSSHTransport,0,192.168.1.105] Connection lost after 25.3 seconds

```

Configuration d'agent Wazuh sur machine honeypot :

```

(kali@kali)-[~]
$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
--2025-12-30 11:09:15-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com) ... 99.84.9.62, 99.84.9.57, 99.84.9.92, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|99.84.9.62|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'

wazuh-agent_4.7  1%[                  ] 126.75K  55.7KB/s

```

```

(kali@kali)-[~]
$ sudo WAZUH_MANAGER='192.168.1.100' WAZUH_AGENT_NAME='agent-vm' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
[sudo] password for kali:
Selecting previously unselected package wazuh-agent.
(Reading database ... 417558 files and directories currently installed.)
Preparing to unpack ... /wazuh-agent_4.7.5-1_amd64.deb ...
Unpacking wazuh-agent (4.7.5-1) ...
Setting up wazuh-agent (4.7.5-1) ...

```



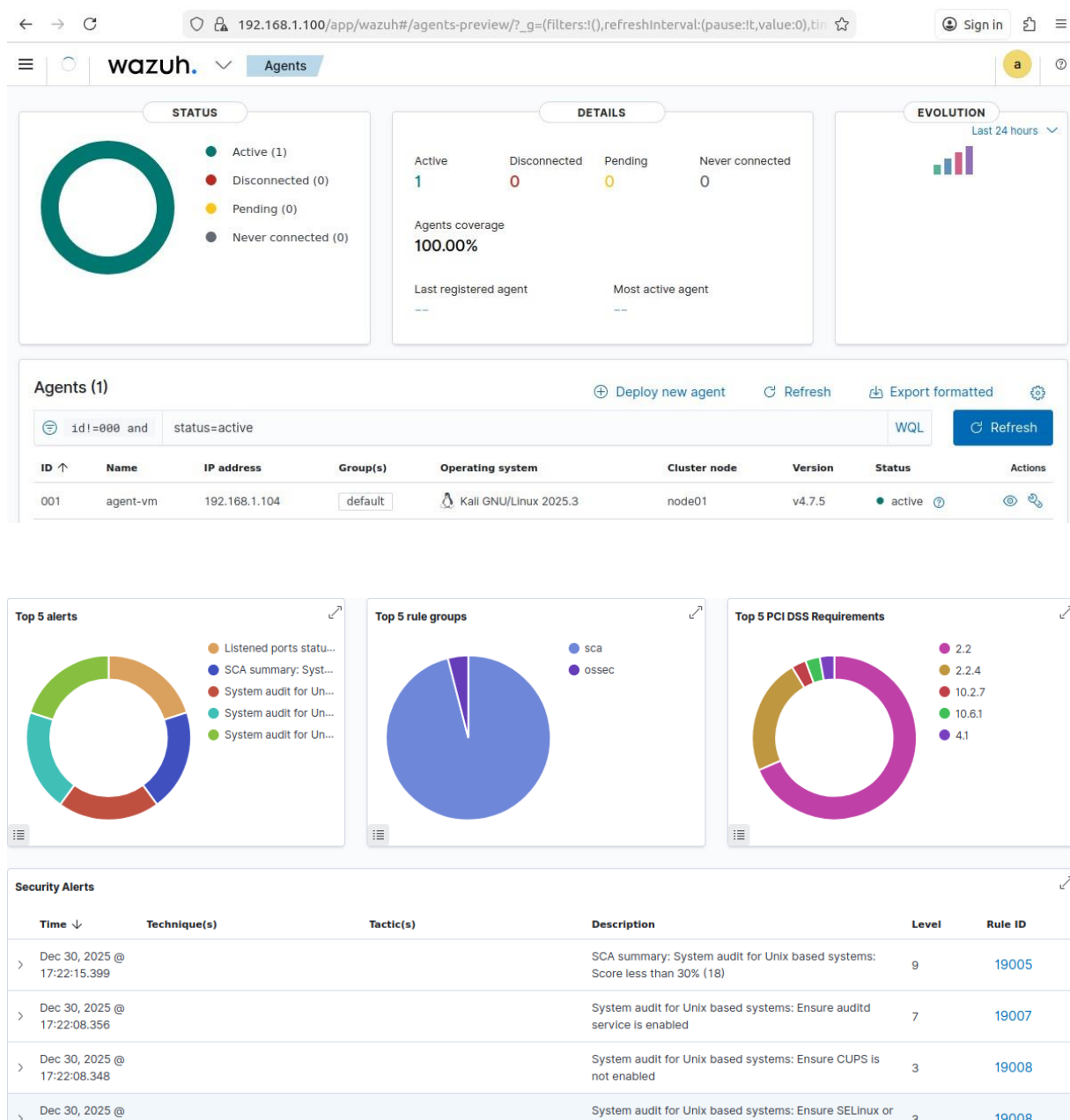
```
(kali㉿kali)-[~]
└─$ sudo systemctl daemon-reload

(kali㉿kali)-[~]
└─$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

(kali㉿kali)-[~]
└─$ sudo systemctl start wazuh-agent
```

```
(kali㉿kali)-[~]
└─$ sudo systemctl status wazuh-agent
[sudo] password for kali:
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; p>
   Active: active (running) since Tue 2025-12-30 11:13:23 EST; 4h 2min ago
   Invocation: cdd3d50d297741f1b3af1cde29f488e7
   Process: 58900 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start>
   Tasks: 32 (limit: 2208)
   Memory: 201.5M (peak: 397.6M)
   CPU: 2min 33.279s
   CGroup: /system.slice/wazuh-agent.service
           └─62984 /var/ossec/bin/wazuh-execd
             └─62992 /var/ossec/bin/wazuh-agentd
               └─63014 /var/ossec/bin/wazuh-syscheckd
                 └─63036 /var/ossec/bin/wazuh-logcollector
                   └─63060 /var/ossec/bin/wazuh-modulesd

Dec 30 11:13:18 kali systemd[1]: Starting wazuh-agent.service - Wazuh agent.>
Dec 30 11:13:18 kali env[58900]: Starting Wazuh v4.7.5 ...
Dec 30 11:13:18 kali env[58900]: Started wazuh-execd ...
Dec 30 11:13:19 kali env[58900]: Started wazuh-agentd ...
Dec 30 11:13:19 kali env[58900]: Started wazuh-syscheckd ...
Dec 30 11:13:20 kali env[58900]: Started wazuh-logcollector ...
Dec 30 11:13:21 kali env[58900]: Started wazuh-modulesd ...
Dec 30 11:13:23 kali env[58900]: Completed.
Dec 30 11:13:23 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.
```



Dans machine soc :

Tu verras que agent est active

```
asmaa@asmaa:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
  ID: 000, Name: asmaa (server), IP: 127.0.0.1, Active/Local
  ID: 001, Name: agent-vm, IP: any, Active

List of agentless devices:
```

- Ajout des chemins de logs dans ce fichier (**machine honeypot**)

```
GNU nano 8.6 /var/ossec/etc/ossec.conf *
<!-- Maximum output throughput -->
<max_eps>50</max_eps>

<!-- Database synchronization settings -->
<synchronization>
  <enabled>yes</enabled>
  <interval>5m</interval>
  <max_eps>10</max_eps>
</synchronization>
</syscheck>

<!-- Log analysis -->
<localfile>
  <log_format>command</log_format>
  <command>df -P</command>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>json</log_format>
  <location>/home/cowrie/cowrie/var/log/cowrie/cowrie.json</location>
</localfile>
```

- Redémarrer l'agent

Sudo systemctl restart wazuh-agent

- Création de règles personnalisées pour Cowrie dans **Machine wazuh**

```
GNU nano 6.2 /var/ossec/etc/rules/cowrie_rules.xml *
<group name="cowrie,honeypot,">
  <rule id="100100" level="6">
    <decoded_as>json</decoded_as>
    <field name="eventid">cowrie.login</field>
    <description>Cowrie: SSH login Attempt</description>
  </rule>
</group>
```


- Redémarre le Wazuh Manager, puis tu trouveras l'alerte



Installer Dionaee via Docker:

- Installation de Docker (si non installé)

```
(kali㉿kali)-[~]
$ sudo apt install -y docker.io docker-compose
sudo systemctl enable docker
sudo systemctl start docker
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0 libgirepository-1.0-1
Use 'sudo apt autoremove' to remove them.

Installing:
  docker-compose  docker.io

Installing dependencies:
  containerd      libintl-perl      libsort-naturally-perl
  criu            libintl-xs-perl   needrestart
  docker-buildx  libmodule-find-perl python3-pycriu
  docker-cli     libnet9           runc
  libcompel1     libproc-processtable-perl tini-static
```

- Téléchargement de l'image Dionaea

```
(kali㉿kali)-[~]
$ sudo docker pull dinotools/dionaea

Using default tag: latest
latest: Pulling from dinotools/dionaea
f22ccc0b8772: Pulling fs layer
3cf8fb62ba5f: Pulling fs layer
e80c964ece6a: Pulling fs layer
e6ff4c28671c: Waiting
e25168386a43: Waiting
```

- Lancement de Dionaea

```
(kali㉿kali)-[~]
$ sudo docker run -d \
  --name dionaea \
  --network host \
  -v /opt/dionaea/log:/opt/dionaea/var/log \
  dinotools/dionaea

999aadaca3806ff4a47033ab4ccd5367cc244d8530d441c9f10a3338c05238fe
```

- Vérification du fonctionnement

```
(kali㉿kali)-[~]
$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED
999aadaca380	dinotools/dionaea	"/usr/local/sbin/ent..."	19 seconds ago

```
Up 18 seconds
dionaea
```

- Test d'attaque (depuis une autre **machine attaquante**)

```
(kali㉿kali)-[~]
$ smbclient -L //192.168.1.104 -N
```

Sharename	Type	Comment
ADMIN\$	Disk	Remote Admin
C\$	Disk	Default Share
IPC\$	IPC	Remote IPC
Printer	Printer	Microsoft XPS Document Writer

```
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.1.104 failed (Error NT_STATUS_CONNECTION_REFUSED)
Unable to connect with SMB1 -- no workgroup available
```

- Journaux (logs) de Dionaea

```
(kali㉿kali)-[~]
$ tail -f /opt/dionaea/log/dionaea/dionaea-errors.log
[31122025 02:03:13] SMB /dionaea/smb/smb.py:779-warning: Attempt to register
12345778-1234-abcd-ef00-0123456789ab failed, TransferSyntax 6cb71c2c-9812-454
0-0300-00000000000000 is unknown
[31122025 02:03:25] SMB /dionaea/smb/smb.py:779-warning: Attempt to register
12345778-1234-abcd-ef00-0123456789ab failed, TransferSyntax 6cb71c2c-9812-454
0-0300-00000000000000 is unknown
[31122025 02:03:34] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1:53
[31122025 02:03:36] SMB /dionaea/smb/smb.py:779-warning: Attempt to register
12345778-1234-abcd-ef00-0123456789ab failed, TransferSyntax 6cb71c2c-9812-454
0-0300-00000000000000 is unknown
[31122025 02:03:44] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1:53
[31122025 02:03:47] SMB /dionaea/smb/smb.py:779-warning: Attempt to register
12345678-1234-abcd-ef00-0123456789ab failed, TransferSyntax 6cb71c2c-9812-454
0-0300-00000000000000 is unknown
[31122025 02:04:27] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1:53
[31122025 02:04:37] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1:53
[31122025 02:05:18] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1:53
[31122025 02:05:28] log_sqlite /dionaea/logsql.py:747-warning: no attackid fo
r 127.0.0.1
```

Signification des Journaux :

Un attaquant tente d'interroger service SMB exposés par le honeypot.

Cela correspond à :

- Du scan SMB
- De l'énumération de services
- Une tentative de détection de vulnérabilités Windows (ex : MS08-067)

Pourquoi ce sont des warnings et pas des errors ?

Dionaea n'est pas un vrai serveur SMB, c'est un honeypot.

- Il simule le service
- Il ne supporte pas tous les protocoles RPC
- Il interrompt volontairement les requêtes complexes

Les warnings indiquent que l'attaque a été détectée et bloquée.

Intégration de Dionaea avec Wazuh

Sur machine honeypot Ajouter la configuration suivante dans l'agent Wazuh dans fichier :
/var/ossec/etc/ossec.conf

Puis redémarrer l'agent

```
GNU nano 8.6 /var/ossec/etc/ossec.conf *
```

```
<localfile>
  <log_format>syslog</log_format>
  <location>/opt/dionaea/log/dionaea/dionaea.log</location>
</localfile>

<localfile>
  <log_format>syslog</log_format>
  <location>/opt/dionaea/log/dionaea/dionaea-errors.log</location>
</localfile>
```

Créer des règles Wazuh pour Dionaea

Par défaut, Wazuh ne comprend pas Dionaea ,il faut des règles.

Sur le Wazuh Manager :

Ajouter une règle personnalisée dans : /var/ossec/etc/rules/dionaea_rules.xml

Puis redémarrer le manager

```
GNU nano 6.2 /var/ossec/etc/rules/dionaea_rules.xml *
```

```
<group name="dionaea,malware,attack">
  <rule id="100200" level="8">
    <match>SMB</match>
    <description>Dionaea - Activity SMB suspecte detecte</description>
    <mitre>T1046</mitre>
  </rule>

  <rule id="100201" level="8">
    <match>Attempt to register</match>
    <description>Dionaea - Tentative d'exploitation SMB</description>
    <mitre>T1210</mitre>
  </rule>

</group>
```

Les alertes sont visibles dans :

- Wazuh Dashboard
- Security Events

timestamp per 30 minutes				
Time ▾	agent.name	rule.description	rule.level	rule.id
> Dec 31, 2025 @ 00:24:31.437	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.414	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.397	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.397	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.396	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.396	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.396	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.395	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.395	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210
> Dec 31, 2025 @ 00:24:31.376	agent-vm	Dionaea - Activity SMB suspecte detecte	8	100210